

20240903 EB_MN (nuclear-cyto-extration)

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- 2024年9月3日 星期二
- split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs
- 2024年9月5日 星期四
- T2 -> T3
- 2024年9月7日 星期六
- T3 -> T4 (~14 mL for over-weekend)
- 2024年9月9日 星期一
- T4 -> T5
- 2024年9月11日 星期三
- T5 -> T6
- 2024年9月13日 星期五
- T6 -> T6 (~14mL for over-weekend)
- 2024年9月16日 星期一
- T6 -> T7
- Dissociation
1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs

2. Remove the PLO and wash the plate with PBS once

3. Coat the plate with laminine for 1hr.

4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).

5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)

7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)

8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

virus: 110 111

Plate A					^
	1	2	3	4	
A	1	2	1	2	
B	3	4	3	4	
C	5	6	7	8	

Plate A conditions					^
	A	B	C	D	
1			treatment	rescue	
2	1	BP 1h			
3	2	Hypo 1h			
4	3	hypo 1h + 1 h rescue			
5	4	hypo 1h + 24h rescue	9/25 7p	9/25 8p	
6	5	BP			
7	6	hypp 1h/3h			
8	7	BP + IPZ			
9	8	hypo + Ver 150nM 3h			

Plate B				^
	1	2	3	
A		hypo	hypo	
B	BP 1h	1h	24h	

2024年9月17日 星期二

Change T7 -> T7 for plate A row A and B

2024年9月18日 星期三

T7-> BP T8 if not added virus

2024年9月20日 星期五

400uL BP T8 -> 500 uL BP T8

2024年9月23日 星期一

400uL BP T8 -> 500 uL BP T8

2024年9月24日 星期二

Treat Plate B A3 and B3 with hypo at 3:30 pm

2024年9月25日 星期三

Plate A

- ☐ B2,4 -> treat hypo at 7pm; rescue at 8pm
- ☐ the rest: 400uL BP T8 -> 500 uL BP T8

Plate B

- ☐ treat A1, B1 with BP T8 at 2:30p
- ☐ treat A2, B2 with hypo at 2:30p
- ☐ collect cells at 3:30p

Extract nucleus with [NE-PER Nuclear and Cytoplasmic Extraction Reagents](#)

Cell Culture Preparation

1. aspirate media
2. wash with ice cold PBS
3. add 1mL ice cold PBS and use lifter to lift the cells gently
4. Transfer 1-10 × 10⁶ cells to a 1.5mL microcentrifuge tube and pellet by centrifugation at 500 × g for 2-3 minutes
5. Use a pipette to carefully remove and discard the supernatant, leaving the cell pellet as dry as possible
6. Add **ice-cold CER I (with protease I)** to the cell pellet (Table 1). Proceed to Cytoplasmic and Nuclear Protein Extraction, using the reagent volumes indicated in Table 1.

This kit is intended for use with fresh (not frozen) cells or tissue samples. Use protease inhibitors to maintain extract integrity and function. Immediately before use, **add protease inhibitors to CER I and NER from concentrated stocks** (e.g., 100X) to minimize reagent dilution. It is **unnecessary to add protease inhibitors to CER II.**

Table1

Table 1. Reagent volumes for different packed cell volumes.*

Packed Cell Volume (μL)	CER I (μL)	CER II (μL)	NER (μL)
10	100	5.5	50
20	200	11	100
50	500	27.5	250
100	1000	55	500

*For HeLa cells, 2 × 10⁶ cells is equivalent to 20μL packed cell volume.

Table1

	A	B	C	D	E	F	G
1		volume/10uL	packed cell volume (uL)	add volume	wells number	total volume (+0.5 well)	protease and phosphatase inhibitor cocktail (100x)
2	CERI	100	10	100	6	650	6.5
3	CERII	5.5	10	5.5	6	35.75	
4	NER	50	10	50	6	325	3.25

Cytoplasmic and Nuclear Protein Extraction

Note: Scale this protocol depending on the cell pellet volume (Tables 1 and 2).

Maintain the volume ratio of CER I:CER II:NER reagents at **200:11:100μL**, respectively.

1. Vortex the tube vigorously on the highest setting for 15 seconds to fully suspend the cell pellet. Incubate the tube on ice for 10 minutes.
2. Add **ice-cold CER II** to the tube.
3. Vortex the tube for 5 seconds on the highest setting. Incubate tube on ice for 1 minute.
4. Vortex the tube for 5 seconds on the highest setting. Centrifuge the tube for 5 minutes at maximum speed in a microcentrifuge (~16,000 × g).
5. Immediately transfer the supernatant (cytoplasmic extract) to a clean pre-chilled tube. Place this tube on ice until use or storage (see Step 10).
6. Suspend the insoluble (pellet) fraction produced in Step 4, which contains nuclei, in **ice-cold NER (with protease I)**.
7. Vortex on the highest setting for 15 seconds. Place the sample on ice and continue vortexing for 15 seconds every 10 minutes, for a total of 40 minutes.
8. Centrifuge the tube at maximum speed (~16,000 × g) in a microcentrifuge for 10 minutes.
9. Immediately transfer the supernatant (nuclear extract) fraction to a clean pre-chilled tube. Place on ice. 10. Store extracts at -80°C until use.